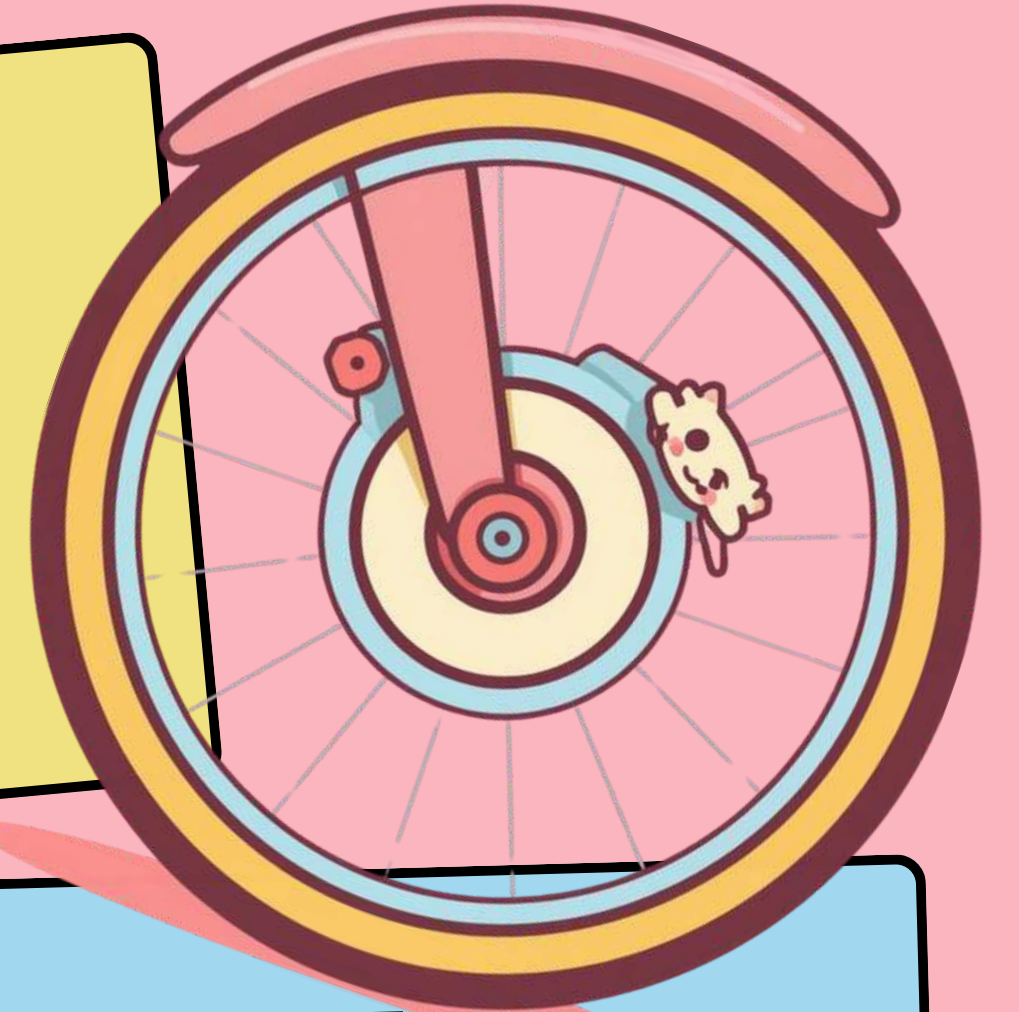


The Stop Secret:



**How Brake Pads Save the
Day**

MEET ISHAAN'S FAST CYCLE!



Ishaan rides his bicycle fast down the street. Suddenly, he sees a big puddle ahead! How will he stop before splashing?

WHAT IS FRICTION? THE INVISIBLE BRAKEMAN

Friction is a tiny invisible force that stops things from sliding. It's like the earth giving your shoes a tiny hug!

- Friction happens when two surfaces rub against each other.
- The rougher the surface, the stronger the grip.



FEEL IT YOURSELF!



I noticed

FRICTION HELPS US EVERY DAY

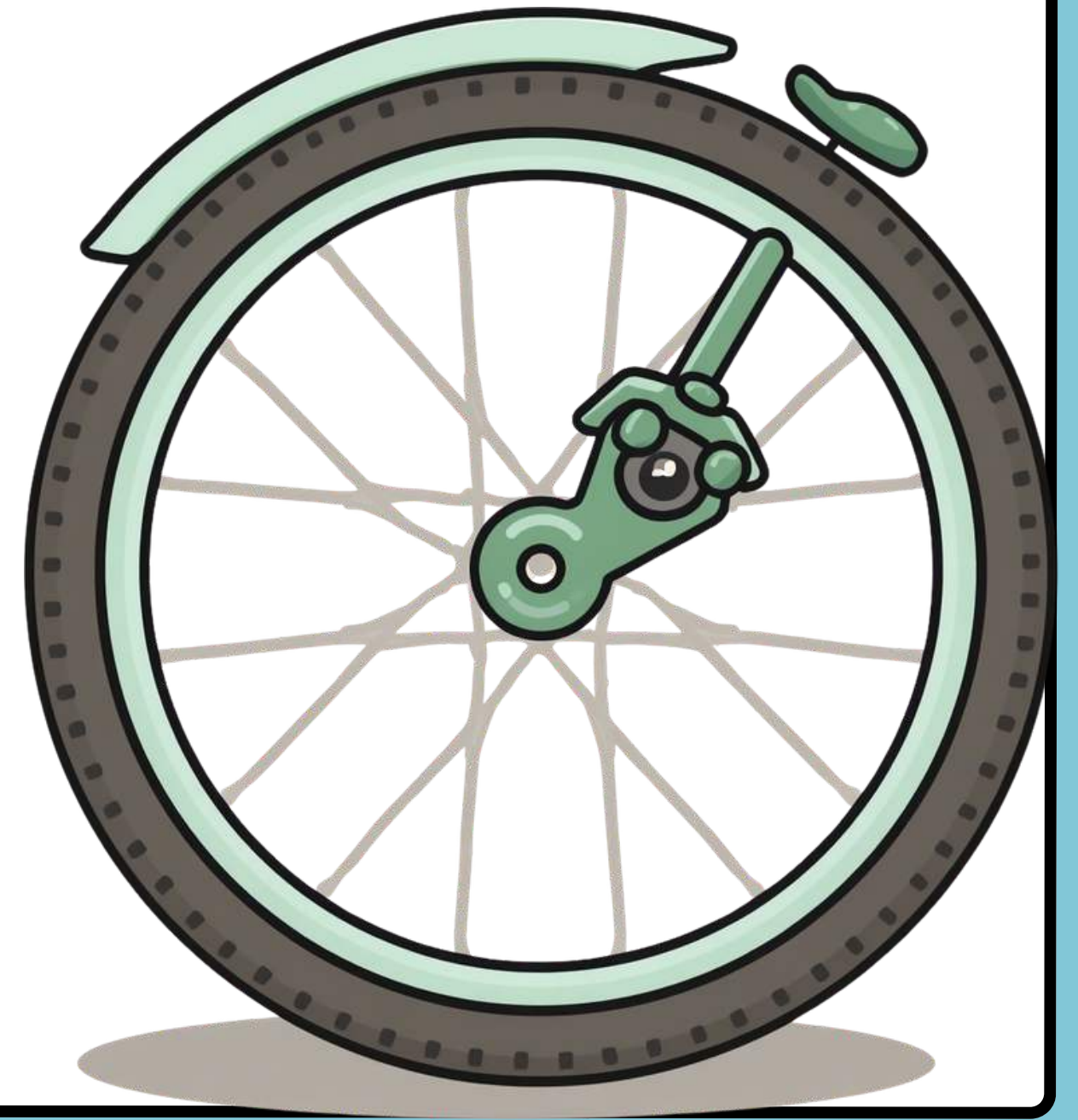
Friction is everywhere around us:

- Walking on rough colony roads wearing Bata chappals—friction stops slipping
- Writing with a pencil gripping paper—friction helps hold the pencil in place
- Playing carrom with talcum powder—less friction helps coins slide smoothly
- Riding a bicycle and stopping safely with brake pads



HOW FRICTION HELPS YOUR BICYCLE STOP

Friction & Brakes



HOW DO BICYCLE BRAKES WORK?



When you pull the brake lever, rubber pads hug the wheel rim. This rubbing creates friction that slows the wheel down.

SEE IT IN SLOW MOTION!



Look closely! The brake pads stop the wheel by rubbing against it.

INVESTIGATION TIME!



Examine the Brake Pads:

- 🔍 What do you see on the brake pad surface?
- 🤔 How do you think the roughness helps stop the bike?

Look closely at the texture and think about friction!

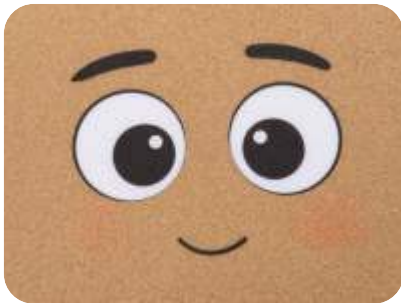
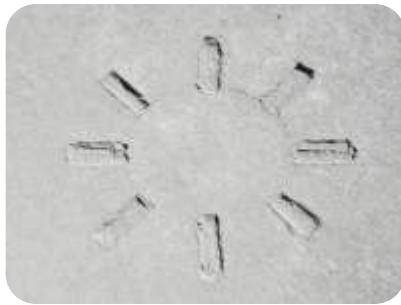
HELPFUL OR HARMFUL? FRICTION IN ACTION

Is friction helpful or harmful? You decide!

- Walking without slipping on rough roads – Helpful!
- Sliding down a water slide – Harmful (we want to slide!)
- Brakes stopping your bicycle safely – Helpful!
- Ice making it hard to walk – Harmful!
- Discuss with your friends: Can you think of more examples?



ROUGH VS SMOOTH: THE CHALLENGE



Rough Surfaces

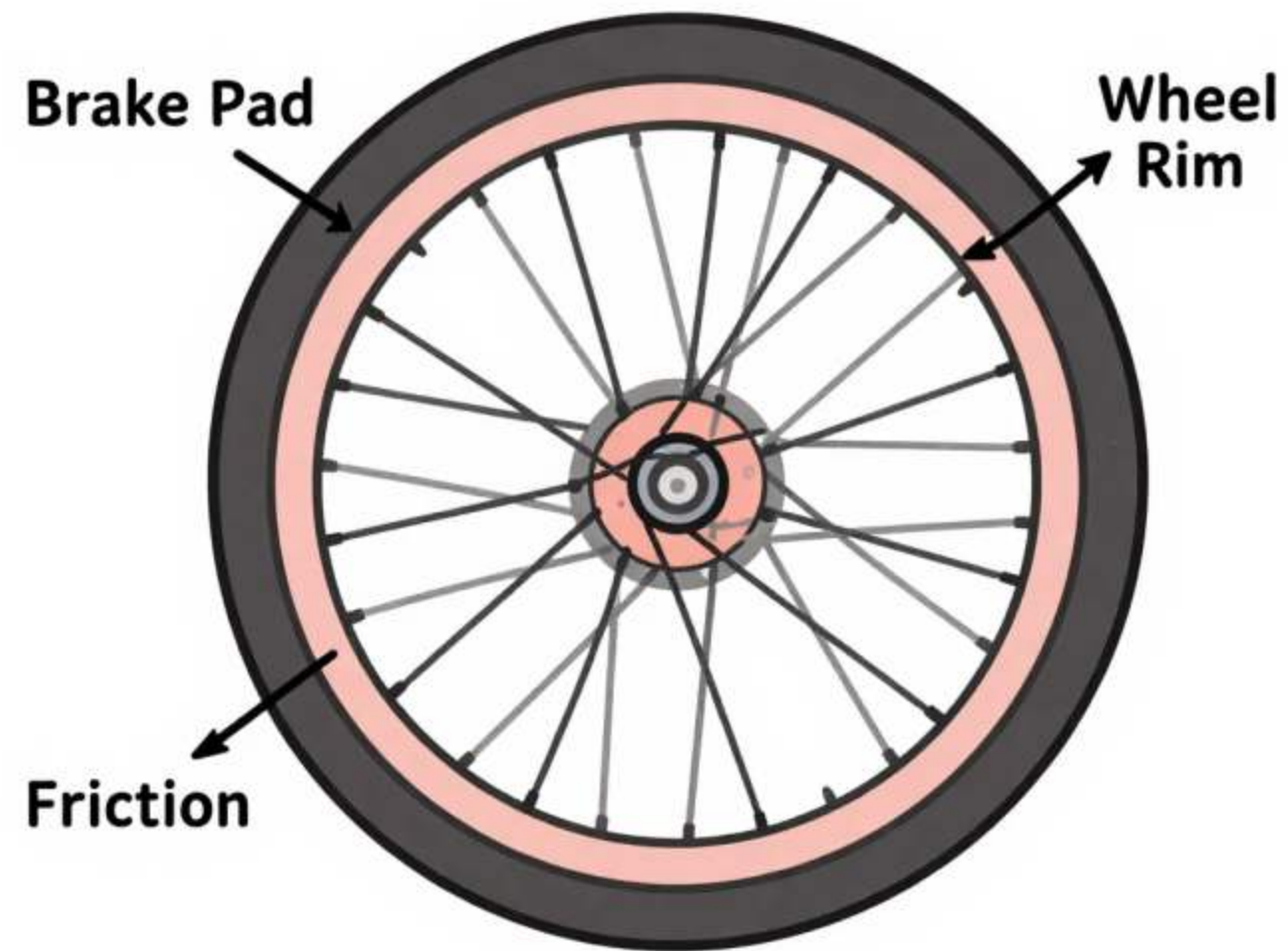


Smooth Surfaces



Which Has More Friction?

THE SECRET OF STOPPING



Friction is the secret force that helps brake pads stop your bicycle safely!

- Brake pads rub against the rim
- This rubbing creates friction
- Friction slows down the wheel
- You stop safely!

SAFETY FIRST!



I noticed:

- Never play with moving bicycle wheels!
- Always be careful near bikes.
- Stay safe and have fun learning!

Thank You!

Any Questions?